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## Motivation

- Uncertainty quantification (UQ)** is essential for reliable ML, yet methods address either **aleatoric** (data noise) or **epistemic** (limited data/model) uncertainty, not both.
- Conformal methods** (e.g., CQR [1]) and **quantile regression** estimate **aleatoric** but **ignore epistemic**, leading to poor **conditional coverage**.
- Ensemble diversity** (e.g., DE [2], PCS [3]) captures **epistemic** but **ignores aleatoric**. DE [2] combines both but with a fixed ratio ( $\lambda=1$ ).
- CLEAR** adaptively balances *both* via two calibration parameters ( $\gamma_1, \lambda$ ).

## Key Idea

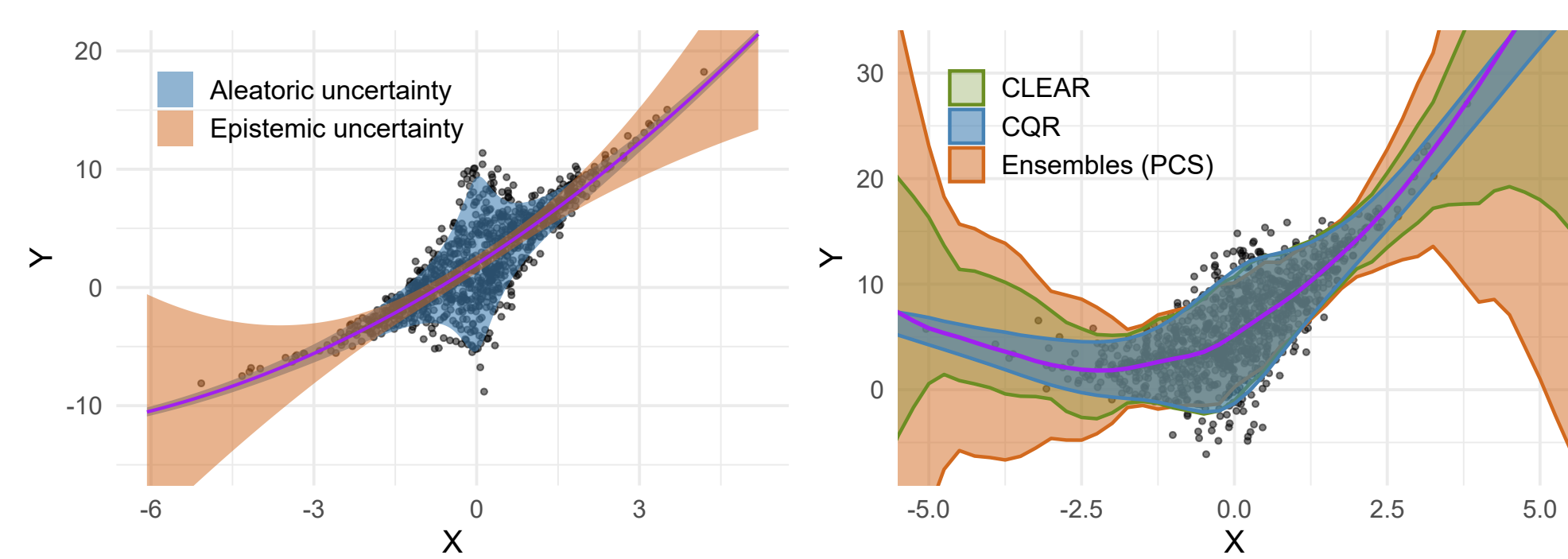


Figure 1: Left: blue = aleatoric, orange = epistemic bands. Right: CQR, PCS, and CLEAR intervals.

$$C(x) = [\hat{f}(x) \pm \left( \frac{\gamma_1}{\text{coverage}} \cdot \text{ale}_{\pm}(x) + \frac{\lambda \gamma_1}{\text{balance}} \cdot \text{epi}_{\pm}(x) \right)]$$

$\hat{f}(x)$ : point prediction,  $\gamma_1$ : ensures **marginal coverage**,  $\lambda$ : **balances** aleatoric vs. epistemic (selected on a held-out set).

- $\lambda = 0$ : only aleatoric;  $\lambda \rightarrow \infty$ : only epistemic
- Adaptive  $\lambda$** : data-driven  $\Rightarrow$  better conditional coverage

## Contributions

- Two calibration parameters** ( $\gamma_1, \lambda$ ) to adaptively balance aleatoric and epistemic uncertainty.
- Residual-based quantile regression** for improved aleatoric estimation.
- First to combine PCS** ensemble perturbation intervals with CQR.
- Large-scale benchmarking** on 17 real-world regression datasets.

## Why CLEAR?

- No retraining**: wraps around any base model
- Model-agnostic**: works with any estimator pair
- Lightweight**: only 2 parameters ( $\gamma_1, \lambda$ )
- Principled**: conformal guarantees + proper scoring

## Theoretical Guarantees

- Marginal coverage**: true value inside the interval  $\geq (1-\alpha)$  of the time, for any finite sample.
- Conditional coverage**: as data grows, coverage holds for each  $x$ , not just on average.
- Proper scoring**: quantile loss ensures honest uncertainty estimates.

### Key References

- [1] Romano et al. *Conformalized quantile regression*. NeurIPS 2019.
- [2] Lakshminarayanan et al. *Deep ensembles*. NeurIPS 2017.
- [3] Yu & Kumbier. *Veridical data science*. PNAS 2020.
- [4] Agarwal et al. *PCS-UQ*. arXiv 2025.
- [5] Tagasovska & Lopez-Paz. *Single-model uncertainties*. NeurIPS 2019.
- [6] Koenker & Bassett. *Regression quantiles*. Econometrica 1978.

## Algorithm

Data is split into training ( $\mathcal{D}_{\text{train}}$ ), calibration ( $\mathcal{D}_{\text{cal}}$ ), and validation ( $\mathcal{D}_{\text{val}}$ ) sets. Target:  $(1-\alpha)$  coverage (e.g.,  $\alpha=0.1$  for 90%).

**Step 1: Estimate epistemic uncertainty** on  $\mathcal{D}_{\text{train}}$

PCS ensembles [3, 4] yield upper/lower epistemic bounds.

**Step 2: Estimate aleatoric uncertainty** on  $\mathcal{D}_{\text{train}}$

Quantile regression on residuals yields upper/lower aleatoric bounds.

**Step 3: Calibrate** on  $\mathcal{D}_{\text{cal}}$  for each candidate  $\lambda$

Find the smallest  $\gamma_1$  so the interval covers  $\geq (1-\alpha)$  of calibration points (conformal guarantee).

**Step 4: Select best  $\lambda^*$  on  $\mathcal{D}_{\text{val}}$**

Pick  $\lambda$  that minimizes quantile loss [6]  $\Rightarrow$  tightest valid intervals.

## Simulation Results

We use a synthetic benchmark to test the interval behavior under controlled noise, reporting calibration and sharpness.

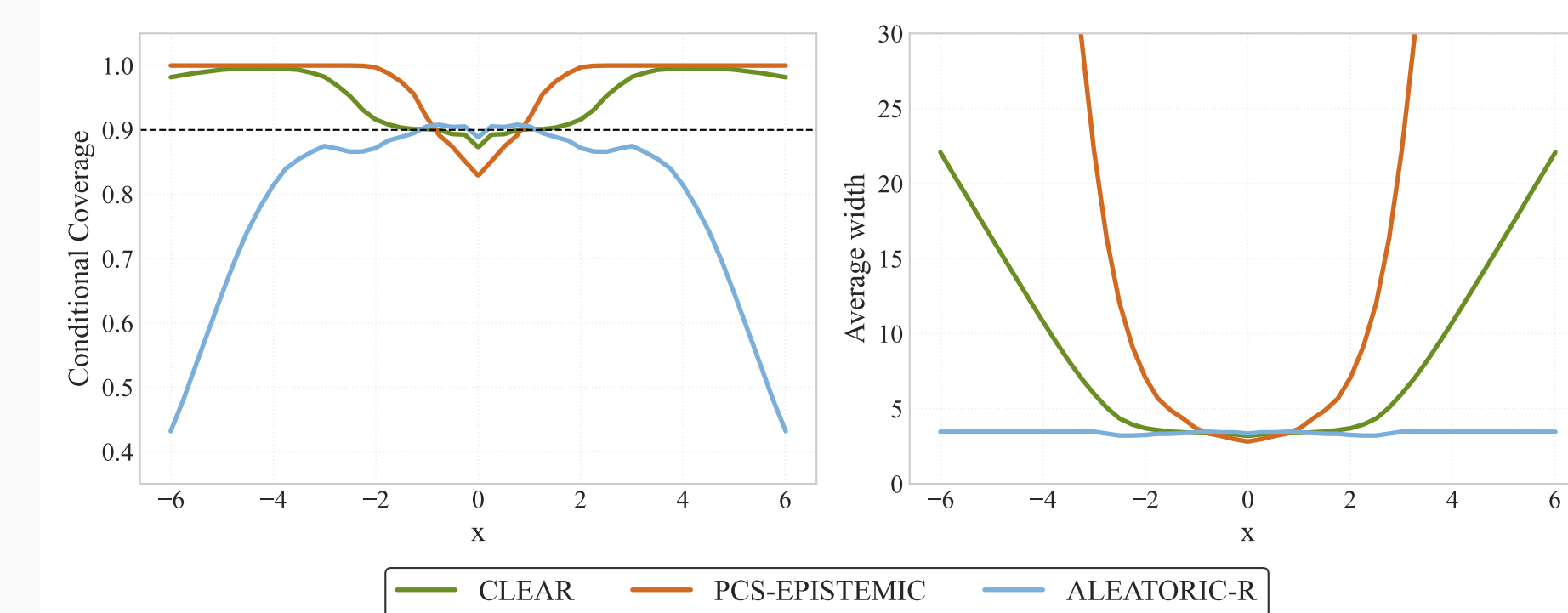


Figure 2: Univariate homoskedastic simulation (100 runs). CLEAR maintains target coverage in extrapolation. Dashed line = 0.9 target.

## Real-World Benchmark: 17 Datasets

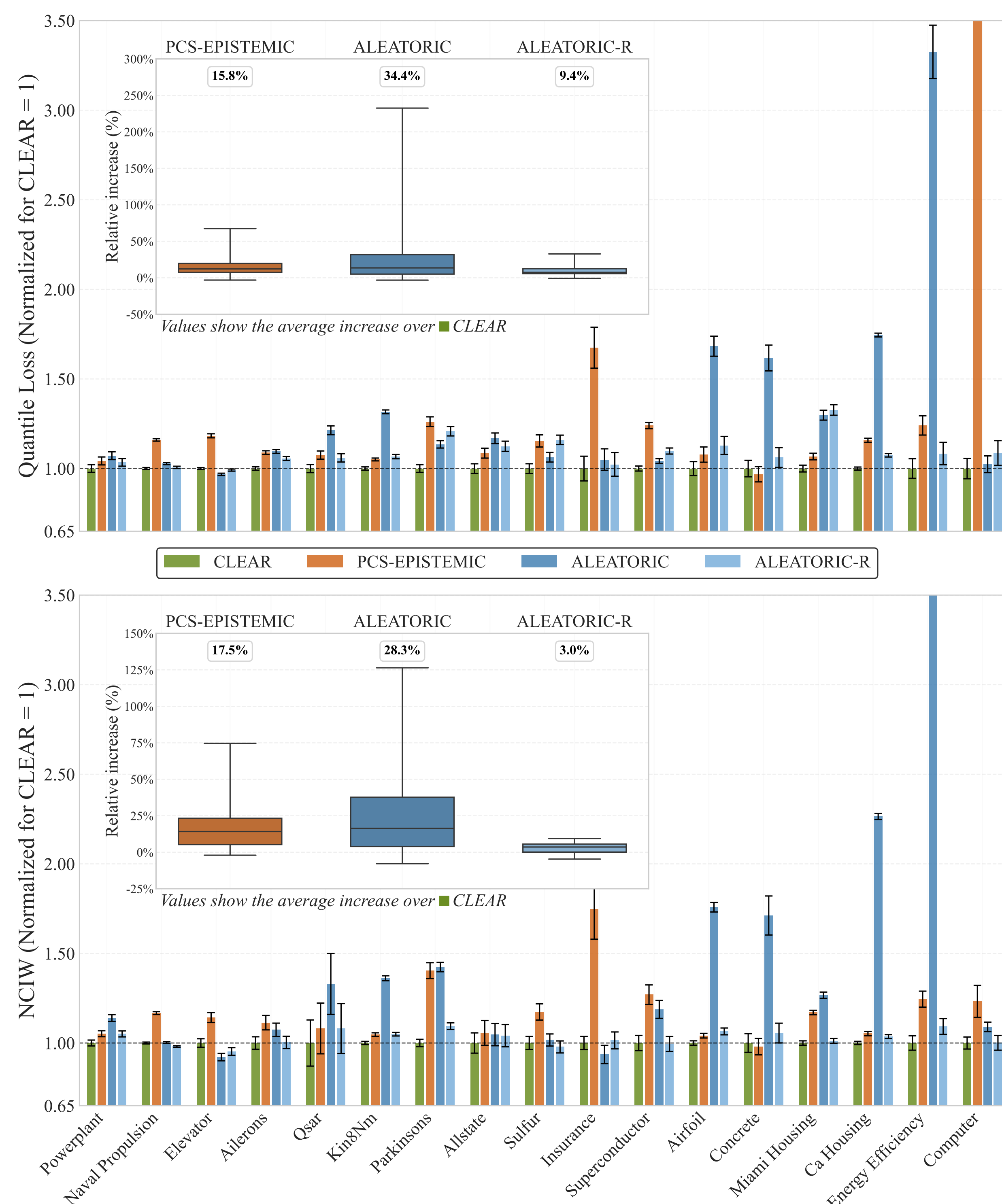


Figure 3: Quantile loss (sharpness) and interval width, normalized to CLEAR (1.0). Lower = better for baselines.

**CLEAR's intervals are tighter than individual parts: +28.2% for aleatoric, +17.4% for epistemic**

Adaptive combination outperforms either component.

## Case Study: Ames Housing

Varying predictors reveals adaptive behavior:

|           | Method       | Width (\$)    | Coverage   | Quantile loss |
|-----------|--------------|---------------|------------|---------------|
| 2 feat.   | PCS          | 107,880       | 87%        | 3,818         |
|           | CQR          | 104,741       | <b>90%</b> | 3,448         |
|           | <b>CLEAR</b> | <b>95,177</b> | 89%        | <b>3,131</b>  |
| All feat. | PCS          | 57,594        | <b>89%</b> | <b>1,922</b>  |
|           | CQR          | 62,398        | 88%        | 2,194         |
|           | <b>CLEAR</b> | <b>55,910</b> | 88%        | 1,923         |

Table 1: Ames Housing (90% coverage target).

- Few feat.  $\Rightarrow$  **aleatoric dominates** ( $\lambda \approx 0.64$ )
- All feat.  $\Rightarrow$  **epistemic dominates** ( $\lambda \approx 14.5$ )
- $\lambda$  **adapts automatically** to the data regime!

## Conclusion

- CLEAR adaptively balances** epistemic & aleatoric uncertainty, outperforming baselines on **17 datasets**.
- Compatible with any pair** of estimators (PCS+CQR, DE+SQR).
- Up to **28.2% tighter prediction intervals** while maintaining valid coverage.
- Theoretical guarantees for **marginal & conditional coverage**.
- Open-source code: `pip install clear-uq`

## Generality: Deep Ensembles + SQR

Also works with DE [2] + SQR [5]:

| Metric                | DE           | SQR          | DE-c.        | SQR-c.       |
|-----------------------|--------------|--------------|--------------|--------------|
| Coverage (PICP)       | +0.05        | −0.66        | −0.09        | −0.15        |
| Interval width (NCIW) | <b>+28.6</b> | <b>+13.4</b> | <b>+27.9</b> | <b>+13.2</b> |
| Sharpness (QL)        | <b>+24.0</b> | <b>+13.7</b> | <b>+24.1</b> | <b>+10.1</b> |

Table 2: Mean % improvement of CLEAR (95% coverage).

$\Rightarrow$  **CLEAR again improves over its own** aleatoric-only and epistemic-only components when using DEs and SQR.